

Scaling Collaborative Filtering with Multimodal Contrastive Fine-tuning

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Abstract

Scaling laws have enabled large language models (LLMs) to achieve remarkable performance and strong generalization across diverse language understanding tasks, including few-shot, in-context, and zero-shot learning. While prior studies in large-scale collaborative filtering (CF) have revealed clear relationships between model performance and scaling factors such as data size and model capacity, little attention has been given to how heterogeneous datasets can be synergistically combined for recommender systems (RS). In particular, it remains unclear whether systematically integrating diverse recommendation datasets can yield scaling behaviors analogous to those observed in LLMs, while simultaneously addressing challenges such as cold-start recommendation and cross-domain transfer. In this paper, we present RecCLIP, a multimodal framework that reformulates user-item interactions as visual representations compatible with vision-language models (VLMs). RecCLIP compresses interaction signals and employs prompt-based ranking to enable unified representation across heterogeneous data sources. Extensive experiments reveal consistent power-law scaling trends with respect to data size, and demonstrate that RecCLIP achieves superior performance in both cold-start and cross-domain transfer scenarios. Our findings underscore the importance of data-centric design in recommender systems and provide practical insights into scaling them effectively. The code for replication is available at <https://github.com/jinliwei-1/RecCLIP>.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Scaling laws, Collaborative Filtering, Large Multimodal Models

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1 Introduction

Scaling laws, which describe empirical relationships between model performance and factors such as model size, data size, and computational capacity, have been extensively validated in natural language processing [19], computer vision [53], and information retrieval [10]. Among these factors, scaling with respect to *data size* has proven particularly critical, as exemplified by GPT-3, which demonstrates that systematically increasing both parameters and training data yields predictable performance gains and enables breakthroughs in few-shot learning through unsupervised pre-training combined with meta-learning [3]. Moreover, scaling laws enable a more generalized learning paradigm, *i.e.*, few-shot learning [44], in-context learning [22], and even zero-shot generalization [33]. These properties are especially appealing for collaborative filtering (CF) [65], which infers user preferences by identifying behaviorally similar users or items and leveraging their collective patterns to generate recommendations. However, this reliance on historical interactions limits CF's ability to adapt to unseen users or items. Consequently, long-standing challenges such as cold-start recommendation [11] and cross-domain transfer [29], which demand robust generalization under sparse or unseen conditions, remain open problems—yet they can be naturally addressed by large-scale models through data scaling. This motivates us to investigate whether scaling laws for CF can be systematically established by integrating and unifying heterogeneous recommendation datasets.

However, unlike LLMs, which can seamlessly integrate training data from heterogeneous sources such as social media conversations, Wikipedia, books, news articles, and web pages [7, 63], scaling up recommender systems from heterogeneous sources remains a non-trivial and largely unresolved task. In particular, the key challenge lies in **unifying representations across heterogeneous sources**. Existing studies have explored the impact of data size

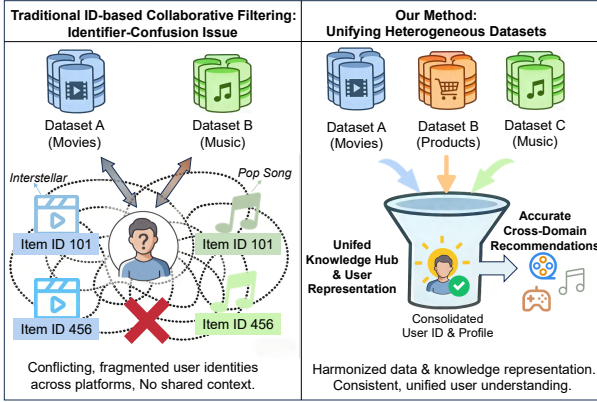


Figure 1: Comparison between traditional ID-based collaborative filtering and our method. The left panel illustrates the identifier-confusion issue encountered in traditional ID-based recommendation, while the right panel demonstrates how our method unifies heterogeneous datasets.

within individual recommendation datasets [46, 54], but they fall short due to the problem of *identifier confusion*, as shown in Figure 1. Typically, users and items in recommendation datasets are represented by arbitrary identifiers. For example, the item ID “101” in a movie dataset may correspond to the film *Interstellar*, while the same ID in a music dataset could represent a pop song. Directly using such raw identifiers obscures semantic distinctions across domains and inevitably confuses the model. In response, recent efforts making use of the LLMs, such as LLM4Rec [27] and semantic-ID [12] methods attempt to enrich identifiers with textual descriptions. Despite these advances, such approaches remain inherently metadata-dependent, and textual descriptions may not always be available due to privacy concerns. This motivates a more general ID-based collaborative filtering framework for unifying heterogeneous recommendation data.

In this work, we propose a new paradigm, termed RecCLIP, as illustrated in Figure 1. By projecting heterogeneous interaction data into a unified image space, our approach enables the model to capture the intrinsic structural reasoning of user-item interactions, independent of domain-specific semantics. This design mitigates identifier confusion and enables scalable, transferable collaborative filtering across diverse domains. Specifically, RecCLIP consists of three components: 1) **Unified visual representation construction**. For each dataset, we convert the interaction histories of the target user and similar users into a unified image representation compatible with vision-language models. As a result, CF signals from different data sources are naturally aligned within a unified space. 2) **Textual prompt construction**. We then transform the target user’s historical interactions into a textual prompt, and append candidate items to the end of the sequence, guiding VLMs toward preference prediction. 3) **Image-text matching**. Finally, we rank candidate items by computing image-text alignment scores between the user interaction image and prompts of candidate items. Through this process, RecCLIP obtains a VLM that can understand and rank items across heterogeneous recommendation datasets in an identifier-agnostic and transferable manner. In summary, the contribution of this paper can be summarized as follows:

- We propose a novel framework, namely RecCLIP, which reformulates collaborative filtering as a vision-language matching problem. By eliminating reliance on dataset-specific identifiers, RecCLIP achieves an identifier-agnostic unified representation across heterogeneous recommendation datasets.
- We conduct extensive experiments to explore the scaling effect against data size across heterogeneous datasets. Our results justify that RecCLIP exhibits a clear scaling law, wherein the test loss and ranking performance vary predictably with data size.
- We further show that RecCLIP enables a generalized learning paradigm, exhibiting strong generalization in both cold-start recommendation and cross-dataset transfer scenarios.

2 Methodology

2.1 Problem Formulation

Let $\mathcal{U} = \{\mathcal{U}^1, \dots, \mathcal{U}^Q\}$ denote the collection of user sets, and $\mathcal{V} = \{\mathcal{V}^1, \dots, \mathcal{V}^Q\}$ denote the collection of item sets aggregated from Q heterogeneous recommendation datasets. Each pair $(\mathcal{U}^q, \mathcal{V}^q)$ represents the user-item space of the q -th dataset, which may differ significantly in domain, interaction patterns, and semantics (e.g., MovieLens, LastFM, Amazon). It is worth noting that this paper focuses on *ID-based CF*, i.e., *learning solely from user-item interaction data without access to any user or item metadata*.

To improve recommendation performance, a reranking process is typically applied to refine the initial recommendation list for each user. Let $\mathbf{V}^r = \{v_1^r, \dots, v_n^r, \dots, v_N^r\}$ denote the initial list containing N items, where $\mathbf{V}^r \subseteq \mathcal{V}^q$ for a given dataset $\mathcal{V}^q$. The reranking procedure aims to generate a more optimal list $\mathbf{V}^{re} = \{v_1^{re}, \dots, v_k^{re}, \dots, v_K^{re}\}$, where $K < N$, by reordering items according to a scoring function $s(u, v)$ that quantifies the preference strength of a user u toward a candidate item v :

$$\mathbf{V}^{re} = \text{Top}_K \underset{v \in \mathbf{V}^r}{s(u, v)}. \quad (1)$$

Therefore, the reranking model refines the initial candidate list $\mathbf{V}^r$ into an optimal recommendation list $\mathbf{V}^{re}$, effectively aligning the ranking with individual user preferences and thereby enhancing the overall recommendation quality. The objective of this work is to learn the scoring function $s(u, v)$ from user-item interactions across Q heterogeneous datasets.

2.2 Unified Visual Representation Construction

Traditional CF methods, such as matrix factorization [20], represent users and items as latent factors within a shared embedding space. Inspired by this idea, we reinterpret the interaction matrix as an image and reformulate CF as a vision-language matching problem. In particular, we construct unified representations by aligning user-item interactions (as images) with candidate items (as textual prompts), enabling effective relevance learning and scalable recommendation across heterogeneous data sources. This section presents our strategy to construct unified representation.

We first construct a binary user-item interaction matrix $\mathbf{M} \in \{0, 1\}^{|\mathcal{U}| \times |\mathcal{V}|}$, where each entry encodes whether a user u has interacted with an item v . The mathematical definition of $\mathbf{M}$ is:

$$\mathbf{M}(u, v) = \begin{cases} 1 & \text{if user } u \text{ interacts with item } v, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

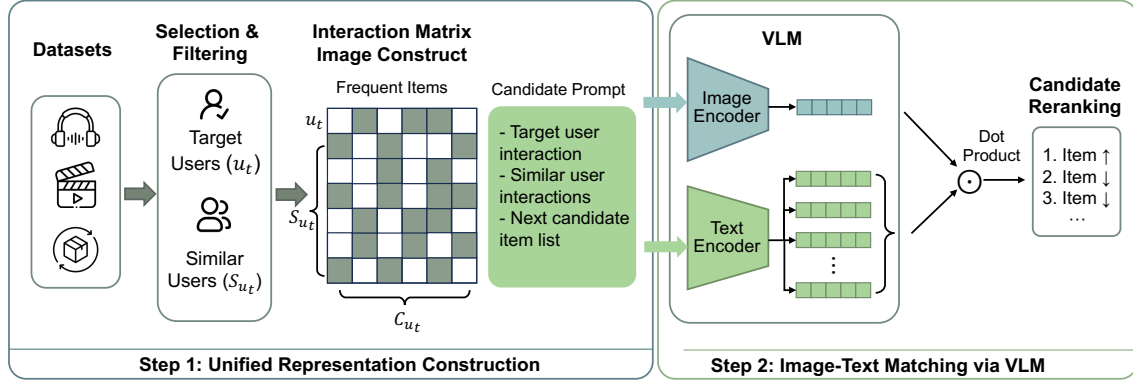


Figure 2: The workflow of RecCLIP. It consists of two steps: (1) constructing images and textual prompts based on target user interactions, which obtains a unified representation across heterogeneous datasets; (2) image-text matching, which leverages the alignment score between encoded features via VLMs for candidate reranking.

This matrix explicitly encodes the sparse interaction patterns between users and items, where entries with value 1 denote observed interactions and entries with value 0 denote unobserved ones.

However, directly leveraging VLMs to process the interaction matrix $\mathbf{M}$ is neither feasible nor efficient. Existing VLMs are typically constrained by limited input resolution, whereas $\mathbf{M}$ can be extremely large and sparse. To address the limitations aforementioned, we design an image construction strategy that converts the interaction patterns of each target user and their similar users into a compact visual representation suitable for VLMs. Given a VLM with a fixed-resolution of $3 \times H \times W$, the follow section introduce our three-step strategy to transform the interaction matrix $\mathbf{M} \in \{0, 1\}^{|\mathcal{U}| \times |\mathcal{V}|}$ to $\mathbf{I}_{u_t} \in \mathbb{R}^{3 \times H \times W}$ for a target user u_t ¹.

2.2.1 Row Selection. Let $u_t \in \mathcal{U}$ denote a target user for whom we aim to predict future interactions. The goal of the row selection strategy is to construct a matrix $\mathbf{R}_{u_t}$ consisting of H rows sampled from the global interaction matrix $\mathbf{M}$ for the target user u_t . Specifically, the first row of $\mathbf{R}_{u_t}$ corresponds to the interaction history of the target user, defined as:

$$\mathbf{r}_{u_t} = \mathbf{M}(u_t, \cdot), \quad (3)$$

where $\mathbf{M}(u_t, \cdot)$ denotes the row of $\mathbf{M}$ associated with u_t . The remaining $H - 1$ rows are filled with the interactions vectors of users who are most similar to u_t . Formally, we measure similarity as:

$$\text{sim}(u, u_t) = \frac{\mathbf{r}_u \cdot \mathbf{r}_{u_t}^\top}{\|\mathbf{r}_u\|_2 \cdot \|\mathbf{r}_{u_t}\|_2}, \quad (4)$$

where $\mathbf{r}_u = \mathbf{M}(u, \cdot)$ is the interaction vector for user u . We then select $H - 1$ users that are most similar to u_t :

$$S(u_t) = \text{Top}_{H-1}(\text{sim}(u, u_t)), \quad u \in \mathcal{U} \setminus \{u_t\}. \quad (5)$$

Finally, we stack the interaction vectors of the target user with those of its most similar users to form the row selection matrix:

$$\mathbf{R}_{u_t} = \begin{bmatrix} \mathbf{r}_{u_t} \\ \mathbf{r}_u \end{bmatrix}_{u \in S(u_t)} \in \{0, 1\}^{H \times |\mathcal{V}|}. \quad (6)$$

¹ H and W are determined by the input resolution of the VLM

This row selection strategy ensures that $\mathbf{R}_{u_t}$ captures both the target user's individual preference patterns and collaborative signals from behaviorally similar users.

2.2.2 Column Selection. To further accommodate the limited input resolution of the vision-language model, we additionally apply a column selection step. It reduces the number of candidate items to a subset that fits within the model's processing constraints. First, we calculate the interaction frequency of each item within the row selection matrix $\mathbf{R}_{u_t}$ for the target user u_t as:

$$f(v) = \sum_{u \in \{u_t\} \cup S(u_t)} \mathbf{R}_{u_t}(u, v), \quad (7)$$

where $f(v)$ denotes the number of users, including u_t and its similar users, who have interacted with item v in $\mathbf{R}_{u_t}$. Afterwards, we select the top W items with the highest interaction frequencies as the candidate item set:

$$C_{u_t} = \text{Top}_W f(v). \quad (8)$$

Finally, we obtain the reduced interaction matrix for the target user u_t by retaining only the selected columns corresponding to C_{u_t} :

$$\mathbf{I}_{u_t} = \mathbf{R}_{u_t}(\cdot, C_{u_t}). \quad (9)$$

Together, the row and column selection processes reduce the dimensions of the original matrix $\mathbf{M}$. This ensures the resulting matrix, $\mathbf{I}_{u_t}$, adheres to the input resolution constraints of the model and is suitable for subsequent visual encoding.

Discussion. In this work, we condense the original interaction matrix $\mathbf{M} \in \{0, 1\}^{|\mathcal{U}| \times |\mathcal{V}|}$ into a compact representation $\mathbf{I}_{u_t} \in \mathbb{R}^{H \times W}$, which is to be converted into image format to serve as the visual input to VLMs. This condensation inevitably introduces a certain degree of information loss, as not all user-item interactions can be preserved within the reduced space. Nonetheless, this compromise represents a pragmatic design choice that balances computational feasibility and representational sufficiency, enabling efficient learning of cross-dataset collaborative patterns within the capacity limits of vision-language models. We leave the exploration for better condense method for future work.

Afterwards, we transform the matrix $\mathbf{I}_{u_t} \in \{0, 1\}^{W \times H}$ into an image representation $\mathbf{I}_{u_t} \in \mathbb{R}^{3 \times H \times W}$, which is compatible with VLMs.

Specifically, we replicate the matrix across three channels and map each entry with value 1 to a black pixel and each entry with value 0 to a white pixel. To this end, we achieve a unified representation of user–item interactions that abstracts away dataset-specific identifiers and enables the integration of heterogeneous recommendation datasets within a shared visual space.

2.3 Preference Inference

This section presents our approach to leveraging the visual representation $\mathcal{I}_{u_t}$ together with a designed textual prompt to enable VLMs to perform user preference prediction. Our idea is to reformulate recommendation as an image–text matching problem, wherein the model evaluates the semantic alignment between the target user’s interaction representation and each candidate item prompts.

2.3.1 Prompt for Candidate Item. To facilitate VLMs in executing reranking tasks, we instantiate a prompt for candidate item $v \in C_{u_t}$ according to the template $\mathcal{P}(C_{u_t}, v)$. Please note that we instantiate the placeholders in the template by *column indices* of the corresponding items, instead of the original identifiers.

This design choice serves two key purposes. First, it eliminates the dependency on dataset-specific identifiers, thereby preventing identifier confusion when aggregating data from heterogeneous sources. Second, it allows the model to focus purely on the relational structure, *i.e.*, collaborative filtering, encoded in the condensed interaction matrix, ensuring that the learned representations remain invariant to the naming or indexing conventions of different datasets. Through this abstraction, the VLM is guided to infer user preferences from the visual patterns in $\mathcal{I}_{u_t}$ while grounding its textual reasoning in the structural roles of the candidate items, rather than in their surface-level semantics.

Candidate Item Prompt Template

The target user history are in the first row, Ids are {Columns indices that the target user has interacted with}.
Row 2 - {H} show the interactions of other users who are similar to the target user.
The ID of the next candidate item is {Candidate Item Index}.

2.3.2 Ranking as Visual–Textual Alignment. To this end, we leverage VLMs to quantify the semantic alignment between the pre-constructed interaction image $\mathcal{I}_{u_t}$ and the textual prompt $\mathcal{P}(C_{u_t}, v)$ corresponding to a candidate item v . Specifically, the VLM serves as the backbone for instantiating the learned relevance function defined in Eq. 1. Formally, we define the matching function as:

$$s(u_t, v) = \text{Score}(\mathcal{I}_{u_t}, \mathcal{P}(u_t, v); \theta), \quad (10)$$

where θ denotes the parameters of the vision–language model. We will introduce the optimization of this relevance function in the next subsection. Here, $\text{Score}(\cdot)$ measures the alignment between the visual embedding of the user’s interaction representation and the textual embedding of the candidate item prompt. Thus, a higher score indicates a higher likelihood of user preference over item v .

After obtaining the matching score $s(u_t, v)$, we utilize it to rerank the items within the candidate item set C_{u_t} . Formally, the final

ranking for the target user u_t is produced by sorting all candidate items $v \in C_{u_t}$ in descending order of their corresponding scores:

$$\mathbf{V}_{u_t}^{re} = \text{Rank}_{\text{desc}}(s(u_t, v)), \quad v \in C_{u_t}. \quad (11)$$

Through this reranking process, items that exhibit stronger visual–textual alignment with the user’s interaction representation are assigned higher ranks, thereby reflecting a higher predicted preference. This step effectively bridges the visual and textual modalities, transforming the matching signal learned by the VLM into a meaningful ranking over candidate items.

2.4 Contrastive learning

Most VLMs are originally designed to capture semantic information from natural images and their textual descriptions, rather than structured interaction patterns. As a result, they are not directly suited for modeling user–item relationships from the interaction matrix $\mathbf{I}_{u_t}$. To bridge this gap, we adopt a contrastive learning paradigm that aligns user–item interaction images with their corresponding textual representations.

Our contrastive loss is designed to maximize the similarity between the visual representation of the target user’s interaction image and the textual representation of a positively interacted item, while simultaneously minimizing the similarity to those of negatively interacted items. Formally, given a target user u_t , a positively interacted item v^+ , and a batch of $\mathcal{B}$ negative items $\{v_1^-, v_2^-, \dots, v_{\mathcal{B}}^-\}$, we define the contrastive loss as:

$$\mathcal{L}(\theta) = -\log \left(\frac{\exp(s(u_t, v^+))}{\exp(s(u_t, v^+)) + \sum_{i=1}^{\mathcal{B}} \exp(s(u_t, v_i^-))} \right) \quad (12)$$

where $s(u_t, v)$ is instantiated in Eq. 10, which represents the alignment score between the visual representation $\mathcal{I}_{u_t}$ and the textual prompt $\mathcal{P}(u_t, v)$. It is worth mentioning that when constructing $\mathcal{I}_{u_t}$, we need to exclude the positive item v^+ from the image to avoid information leakage, *i.e.*, the column corresponding to the positive item needs to be converted into white pixels.

This loss encourages the model to assign higher scores to positively interacted items while penalizing negative ones, thereby driving the learning of a discriminative relevance function capable of generalizing across heterogeneous datasets.

3 Experiments

In this section, we conduct extensive experiments to evaluate the effectiveness of our proposed method. Specifically, our experiments are designed to address the following research questions:

- **RQ1:** Can RecCLIP exhibit scaling laws with respect to data size on heterogeneous recommendation datasets, analogous to those observed in LLMs?
- **RQ2:** How does RecCLIP perform compared to state-of-the-art baselines on the reranking task?
- **RQ3:** How does the generalization ability of RecCLIP compare with that of collaborative filtering (CF) methods, as well as methods specifically designed for cold-start and transfer learning scenarios—particularly in these two scenarios?
- **RQ4:** What are the contributions of the individual components of RecCLIP to the overall performance?

Table 1: Statistics of the datasets.

Datasets	LastFM-2K	MovieLens-1M	Amazon Games
# Users	1,143	6,040	50,547
# Items	11,854	3,706	16,859
# Interactions	68,436	1,000,209	389,718
# Interactions / # user	59.92	165.59	7.71
# Interactions / # item	5.77	269.88	23.11

3.1 Experimental Setups

3.1.1 Datasets. To explore the generalization of the proposed model, we evaluate our approach on three widely recognized datasets: LastFM-2K [4], MovieLens-1M [15] and Amazon Games [31]. The statistics of these datasets are summarized in Table 1. We preprocess the data using 5-core filtering [16, 41] and evaluate our approach under the leave-one-out protocol [13].

3.1.2 Baselines. To comprehensively evaluate our method, we compare against the following baseline methods: **GMF** [17], **DLCM** [2], **BM25** [35], **NGCF** [41], **LightGCN** [16], **DGCF** [42], **UltraGCN** [30], **RankGPT** [37], **PNN** [60], and **TIGER** [34]. For the specialized cold-start and transferable recommendation scenarios, we include **CLCRec** [45] and **TransRec** [38]. Detailed descriptions of all baselines are provided in the Appendix B.

3.1.3 Experimental Protocols. Our method is initialized with a pre-trained VLM. In this paper, we use the CLIP model as the backbone. To verify the general applicability of our method, we also used the BLP [21] model as the backbone to conduct experiments. In the training and testing phases, we set the number of negative samples $\mathcal{B}$ to 10. Training is conducted on 4 GPUs using a micro-batch size of 2 and 8 gradient accumulation steps, yielding a total batch size of 64. We use the AdamW optimizer with a learning rate of $5e-6$ and a weight decay of $1e-3$. We train for 10 epochs and adopt an early stopping mechanism. Further details can be found in Appendix C.

To evaluate the ranking performance, we adopt the widely used metrics Recall@K (R@K) and NDCG@K (N@K). For both metrics, we report the results at ranks 1, 3, and 5 to show the performance of models on different positions, and all metrics are reported in percentages (%). Statistical differences are computed based on the paired t-test with $p \leq 0.05$.

3.2 Data Size Scaling (RQ1)

This section investigates how data size influences model performance across heterogeneous datasets. The training data size, measured by the number of user-item interactions, is systematically varied across 14K (the full LastFM training set), 60K, 107K, 153K, 200K, 246K, and 293K samples, respectively. We evaluate each fine-tuned model on the LastFM test set to assess scaling laws with increasing data size.

Our experimental results are presented in Figure 3. While performance consistently improves with more data, the gains diminish, indicating that the model’s performance progressively approaches saturation. As Figure 3 illustrated, increasing the data size leads to a gradual decrease in test loss and a consistent improvement in NDCG@5. In particular, we model the scaling law with respect to data size. The test Loss L is assumed to follow an inverse power law approaching an irreducible loss floor E_{loss} :

$$L(N) = A \cdot N^{-\alpha} + E_{\text{loss}}, \quad (13)$$

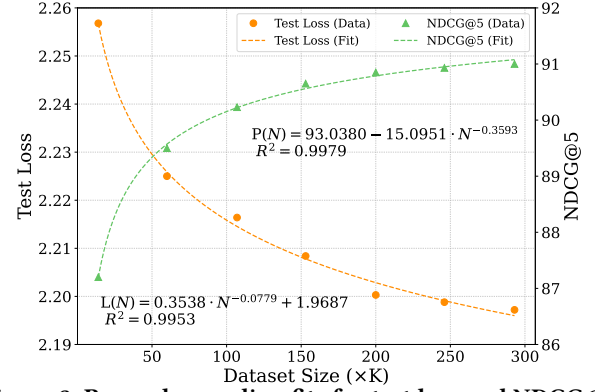


Figure 3: Power-law scaling fits for test loss and NDCG@5. The markers represent observed empirical results, while the dashed curves denote the best-fit lines obtained from the inverse power-law model.

where N represents the number of user-item interactions in the training set, $L(N)$ denotes the model’s test loss, parameters A and α are the scaling coefficients, and E_{loss} is the asymptotic error as $N \rightarrow \infty$. Correspondingly, the performance metric P is modeled as a saturating function approaching a performance ceiling E_{perf} :

$$P(N) = E_{\text{perf}} - B \cdot N^{-\beta}, \quad (14)$$

where $P(N)$ denotes the NDCG@5 score on the test set, parameters B and β are the scaling coefficients, and E_{perf} represents the model’s maximum achievable performance. In both cases, the coefficient of determination (R^2) indicates a good fit. Based on these results, we infer that the test loss and NDCG@5 score follow a power-law scaling relative to the data size. This finding justifies that **RecCLIP is able to exhibit scaling laws with respect to data size on heterogeneous recommendation datasets.**

3.3 Overall Performance (RQ2)

In this section, we conduct a comprehensive comparison between RecCLIP and various baselines. Our results are summarized in Table 2, and we have the following observations.

RecCLIP significantly outperforms all baseline methods when optimized with heterogeneous datasets. This result demonstrates the effectiveness of RecCLIP for real-world recommender systems. Collaborative filtering is well known to suffer from data sparsity, which limits its ability to capture complex user-item relationships. In contrast, RecCLIP alleviates this limitation by leveraging heterogeneous datasets, enriching the training signal and improving recommendation performance. The observed gains stem from unifying heterogeneous data, which enables the model to learn richer and more comprehensive representations.

RecCLIP achieves superior performance without relying on item metadata. Existing methods, such as RankGPT and TIGER, require access to item semantic information for effective generalization, whereas RecCLIP outperforms them across all evaluation metrics using interaction data alone. This independence from semantic features is a key advantage of RecCLIP, particularly in settings where item metadata is sparse, unavailable, or restricted due to privacy concerns. We leave the integration of item metadata with RecCLIP for future work.

Table 2: Overall performance comparison between RecCLIP and baseline methods on LastFM-2K, MovieLens-1M, and Amazon Games. The best and second-best results in each column are highlighted in bold and underlined. “*” indicates statistically significant improvement over the best baseline with p -value ≤ 0.05 .

Method	LastFM-2K						MovieLens-1M						Amazon Games					
	R@1	R@3	R@5	N@1	N@3	N@5	R@1	R@3	R@5	N@1	N@3	N@5	R@1	R@3	R@5	N@1	N@3	N@5
GMF	63.45	72.18	76.54	63.45	68.32	70.15	78.23	83.56	87.12	78.23	81.45	82.98	75.12	81.34	85.67	75.12	78.90	80.45
DLCM	65.80	74.20	78.10	65.80	70.50	72.80	80.15	85.90	88.45	80.15	83.40	84.90	77.30	83.10	86.80	77.30	80.50	82.10
BM25	58.76	66.54	70.23	58.76	62.12	64.56	72.34	78.56	82.12	72.34	75.67	77.89	70.45	76.23	80.12	70.45	73.56	75.23
NGCF	71.23	81.56	85.45	71.23	74.89	77.23	85.67	89.45	92.34	85.67	88.12	90.45	82.34	87.56	91.23	82.34	86.45	89.12
LightGCN	74.99	85.12	87.87	74.99	77.15	80.77	87.90	91.78	94.60	87.90	90.86	93.52	85.35	89.89	94.28	85.35	89.01	92.47
DGCF	76.23	86.45	89.12	76.23	78.56	82.34	88.56	93.23	95.45	88.56	91.45	93.63	86.78	91.23	95.12	86.78	90.34	93.12
UltraGCN	<u>77.45</u>	<u>87.12</u>	89.56	<u>77.45</u>	79.89	83.12	<u>89.12</u>	<u>94.56</u>	<u>96.23</u>	<u>89.12</u>	<u>92.34</u>	<u>93.89</u>	87.45	92.12	95.89	87.45	91.56	93.78
RankGPT	66.12	74.55	78.35	66.12	70.85	73.05	80.45	86.12	88.78	80.45	83.65	85.12	77.65	83.45	87.15	77.65	80.89	82.45
TransRec	73.45	83.12	86.50	73.45	76.10	79.25	86.12	89.85	92.80	86.12	88.45	90.95	84.85	89.15	93.45	84.85	88.20	91.50
CLCRec	73.80	83.50	86.85	73.80	76.45	79.60	86.55	90.25	93.10	86.55	88.80	91.20	84.20	88.65	92.90	84.20	87.80	91.10
PNN	69.45	79.10	82.30	69.45	73.50	75.80	88.65	92.15	94.90	88.65	91.20	93.85	81.20	86.40	90.10	81.20	85.10	87.50
TIGER	77.20	86.50	89.80	77.20	83.10	84.90	86.50	90.80	93.90	86.50	89.40	91.50	<u>88.10</u>	<u>94.20</u>	<u>96.50</u>	<u>88.10</u>	<u>92.50</u>	<u>93.80</u>
RecCLIP	80.65*	89.75*	92.29*	80.65*	86.18*	87.20*	92.53*	98.46*	98.86*	92.53*	96.21*	96.37*	92.36*	97.98*	98.34	92.36*	95.24*	95.70

Table 3: Cold-start recommendation performance comparison between RecCLIP and baseline methods on LastFM-2K, MovieLens-1M, and Amazon Games. The best and second-best results in each column are highlighted in bold and underlined. “*” indicates statistically significant improvement over the best baseline with p -value ≤ 0.05 .

Method	LastFM-2K						MovieLens-1M						Amazon Games					
	R@1	R@3	R@5	N@1	N@3	N@5	R@1	R@3	R@5	N@1	N@3	N@5	R@1	R@3	R@5	N@1	N@3	N@5
NGCF	35.00	55.33	66.67	35.00	47.00	51.66	35.85	69.04	83.50	35.85	55.22	61.21	41.00	68.95	81.35	41.00	57.23	62.33
LightGCN	41.33	58.00	72.33	41.33	50.80	56.64	37.47	69.65	86.15	37.47	56.18	62.97	47.20	73.59	84.36	47.20	62.60	67.04
DGCF	45.00	66.67	79.00	45.00	57.54	62.61	43.18	75.36	85.95	43.18	61.96	66.35	57.43	80.48	88.72	57.43	70.96	73.47
UltraGCN	36.00	55.67	64.00	36.00	47.54	50.93	35.23	70.06	85.13	35.23	55.45	61.73	40.91	68.88	81.90	40.91	57.20	62.56
CLCRec	<u>63.67</u>	<u>81.33</u>	<u>84.00</u>	<u>63.67</u>	<u>74.33</u>	<u>77.03</u>	<u>84.99</u>	<u>88.58</u>	<u>90.91</u>	<u>84.99</u>	<u>88.22</u>	<u>90.25</u>	<u>81.47</u>	<u>86.22</u>	<u>90.71</u>	<u>81.47</u>	<u>85.75</u>	<u>87.30</u>
RecCLIP	70.33*	93.00*	97.33*	70.33*	83.89*	85.67*	89.41*	99.59*	99.8*	89.41*	95.62*	95.7*	89.94*	98.77*	98.83*	89.94*	94.16*	94.19*

3.4 Generalization Analysis (RQ3)

3.4.1 Cold Start Recommendation. In this section, we investigate the how RecCLIP addresses the cold-start problem. Following prior work [58], we categorize the users in each dataset into two groups: regular users and cold-start users, based on their number of interactions, determined by a threshold hyperparameter T . Specifically, T is set to 10 for LastFM, 25 for MovieLens-1M, and 5 for Amazon Games. For the cold-start experiments, we adopt NGCF, LightGCN, DGCF, UltraGCN, and CLCRec as baseline models, particularly CLCRec is designed for cold-start scenarios.

The experimental results for cold-start setting are summarized in Table 3. **Our method consistently achieves the best performance across all metrics and datasets.** For example, on the Recall@3 metric, RecCLIP outperforms the strongest baseline by 14.3%, 12.4%, and 14.6% on the LastFM-2K, Amazon Games, and MovieLens-1M datasets, respectively.

These substantial gains highlight the superior generalization capability of RecCLIP under sparse user interaction settings. In summary, the results demonstrate that our method is well-equipped to handle cold-start scenarios and exhibits strong robustness against data sparsity.

3.4.2 Transferable Recommendation. This section investigates the transferability of RecCLIP across heterogeneous datasets. We train each model on a single source dataset and then evaluate its performance on two unseen *target* datasets. We compare RecCLIP to four competitive GNN-based models, including NGCF, LightGCN,

DGCF, and UltraGCN, along with TransRec, a model specifically designed for transferable recommendation. For clarity, Figure 4 illustrates the results of all methods trained on the LastFM dataset and tested on the MovieLens-1M and Amazon Games datasets. The comprehensive results are provided in Appendix D. The results lead to the following observations.

RecCLIP demonstrates strong transferability, effectively generalizing to unseen domains. As shown in Figure 4, our proposal consistently and significantly outperforms all baseline models across all datasets and evaluation metrics. In addition, **data scale is a key factor influencing the manifestation of the scaling law.** As shown in Figure 5, RecCLIP achieves better performance when the model is trained on the MovieLens-1M or Amazon Games datasets and tested on LastFM. This discrepancy can be attributed to the larger data volumes of MovieLens-1M and Amazon Games, which provide richer interaction signals for learning transferable representations. **The interaction density of the data sources also plays a crucial role in transferability.** As shown in Table 1, the MovieLens-1M dataset exhibits a substantially higher user interaction density than LastFM, offering richer collaborative signals that facilitate stronger cross-domain generalization.

The above observations provide a principled strategy for leveraging RecCLIP: **use large-scale, high-interaction datasets as foundation datasets to pretrain the model, and then fine-tune it on the target dataset, even when the latter is relatively small in scale.**

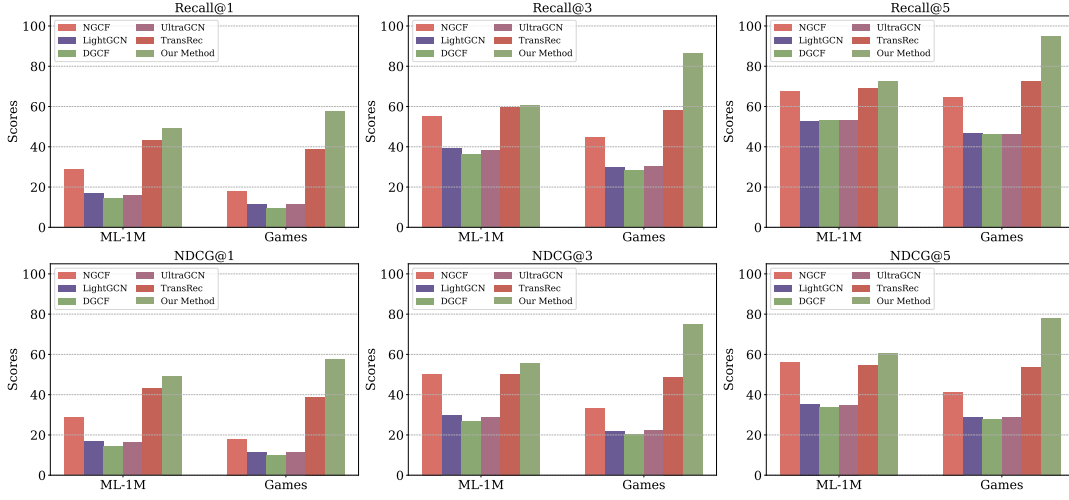


Figure 4: Transferable recommendation performance comparison between RecCLIP and baselines on the MovieLens-1M and Amazon Games with different evaluation metrics, where all models are trained on the LastFM-2k dataset.

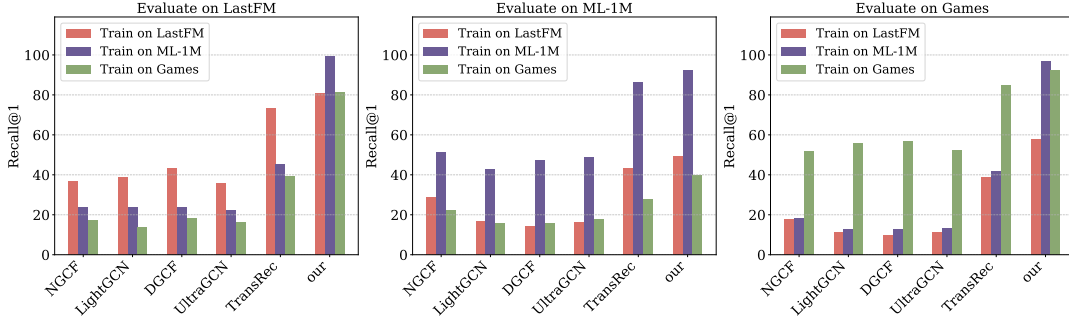


Figure 5: Transferable recommendation Recall@1 performance under in-domain and cross-domain training, highlighting the effect of data scale on cross-domain generalization.

Table 4: Ablation study on VLM backbones. The performance between alternatives is not significant with p -value ≤ 0.05 .

Method	LastFM-2K					
	R@1	R@3	R@5	N@1	N@3	N@5
CLIP	80.65	89.75	92.29	80.65	86.18	87.2
BLIP	79.84	88.96	92.11	79.84	85.47	87.52
Method	MovieLens-1M					
	R@1	R@3	R@5	N@1	N@3	N@5
CLIP	92.53	98.46	98.86	92.53	96.21	96.37
BLIP	93.12	98.22	98.91	93.12	96.85	96.42
Method	Amazon Games					
	R@1	R@3	R@5	N@1	N@3	N@5
CLIP	92.36	97.98	98.34	92.36	95.24	95.7
BLIP	91.78	98.06	98.15	91.78	95.58	95.41

3.5 Ablation Study (RQ4)

Our ablation study investigates the influence of the choice of backbone architecture and the column selection strategy.

3.5.1 Backbone Analysis. To investigate the general applicability of our method across different backbones, we compare CLIP with

Table 5: Ablation study of column selection strategy on LastFM-2K. “*” indicates statistically significant improvement over the baseline with p -value ≤ 0.05 .

Method	LastFM-2K					
	R@1	R@3	R@5	N@1	N@3	N@5
Random	22.50	43.17	59.19	22.5	34.2	40.77
Frequent	80.65*	89.75*	92.29*	80.65*	86.18*	87.20*

BLIP. We use BLIP with the blip-itm-coco variant. Experimental results are shown in Table 4. The results demonstrate that the performance differences among various VLM backbones are not statistically significant. This indicates that the core mechanism of RecCLIP can effectively adapt to different pre-trained models, further validating the general applicability of our approach.

3.5.2 Column Selection Strategy. We compare different column selection strategies in the image construction process. Specifically, we replace the frequency-based column selection with a random column selection strategy to assess its impact on performance. The experimental results are presented in Table 5.

From the experimental results, we observe that the performance metrics drop sharply when columns are selected at random. This finding confirms the effectiveness of our high-frequency column

selection strategy, which maximizes the retention of interaction information and maintains sufficient interaction density.

4 Related Work

4.1 Large Model-based Recommendation

LLM-based Recommendation leverages the text understanding, generation, and reasoning capabilities of LLMs to enhance the personalization, interpretability, and efficiency of recommender systems [6, 49, 62]. Existing methods explore automatic content generation [1], semantic representation learning [24], and the discovery of latent relational structures for sequential recommendation [48]. Specially designed prompts are also used to exploit the contextual learning abilities of LLMs for zero-shot recommendation [39]. More recently, generative recommendation systems have emerged as a promising paradigm [9]. These methods learn to map item to semantic ID according to item metadata, and use generative retrieval to directly predict the semantic ID of the next item.

Given that LLMs mainly rely on textual input and may contain factual errors [61], recent studies have explored VLMs to incorporate richer user-item interaction signals. VLM-based recommendation leverages these models to process and integrate diverse data types such as text, images, and videos, enabling more comprehensive user and item representations and addressing the limitations of traditional text-based recommender systems [18, 25, 28, 57]. Despite these advances, existing LLM- and VLM-based recommendation methods share a common limitation: a strong dependence on item- or user-side metadata, such as textual descriptions, images, or attributes, to enable generalization and cross-domain transfer. To the best of our knowledge, this work is the first to investigate the use of VLMs to directly scale from raw user-item interaction patterns alone, without relying on external metadata, thereby enabling a unified representation for cross-domain recommendation.

4.2 Transferable and Cold-start Recommendation

Transferable recommendation aims to enhance model performance in new domains by leveraging knowledge across datasets, addressing data sparsity or cold-start issues in scenarios like e-commerce and social media [50, 52]. Early methods relied on explicit overlaps and mapping functions [55, 59], while recent efforts such as RUNSRec [56] attempt to learn universal representations using textual information and contrastive learning. However, inconsistent user and item identifiers across domains remain a challenge, limiting the generalization ability of the dominant IDRec paradigm [51]. Nonetheless, domain-specific discrepancies hinder truly unified modeling [36], and scaling laws, though promising, have not been fully validated in recommendation settings.

The cold start problem, where systems fail to generate quality recommendations for new users or items due to insufficient data, remains a critical challenge with direct impacts on user engagement and business growth [23, 64]. To address this, the research community has developed various approaches. For example, Cohen et al. [8] proposed mapping item attributes to latent spaces via a learned Gaussian distribution. Hao et al. [14] introduced MPT, a more general pre-training framework that enhances a Graph Neural Network with a transformer and contrastive learning to better

leverage node features. Recently, in sequential recommendation, CMCLRec [47] tackled user cold start by using cross-modal contrastive learning to generate synthetic behavioral sequences from user features. Despite these advances, existing methods share a common reliance on auxiliary semantic features. This dependency, however, introduces a fundamental bottleneck in practice: such side information is frequently unavailable, whether due to stringent privacy regulations or users' own data-sharing preferences. This severely limits real-world applicability, thereby creating a need for methods that operate effectively without external supervision, a need that this paper directly addresses.

4.3 Scaling Law in Recommendation

Scaling laws describe that performance improves with scale, successfully applied in language modeling and graph-based recommendation [43]. Current efforts to harness these laws follow three main paradigms: (1) designing self-supervised tasks, often using generative models like GANs or diffusions to augment sparse data [5, 40]; (2) leveraging large-scale language or multimodal models to process item content via Transformer architectures [26, 40]; and (3) integrating recommenders as tools within LLM agent frameworks for personalization [39]. However, current scaling laws for recommender systems are limited by data sparsity. To address this, we propose a unified multimodal method aligning visual and textual data via contrastive learning. This approach models user-item interactions from a multimodal perspective, improving scalability and cross-domain generalization while adhering to scaling laws.

5 Conclusion and Future Work

In this work, we propose RecCLIP, a novel paradigm that establishes the scaling law for collaborative filtering using heterogeneous datasets. We reformulate collaborative filtering as a vision-language matching problem, thereby addressing the issue of identifier dependence and enabling an identifier-agnostic representation space. In particular, RecCLIP converts user-item interactions into visual and textual modalities, unifying heterogeneous recommendation datasets under a shared multimodal framework and laying the foundation for the emergence of scaling laws in recommendation. Extensive experiments demonstrate that RecCLIP exhibits predictable scaling laws with respect to data size, shows strong generalization in cold-start and cross-domain transfer scenarios, and maintains robustness across different backbone architectures. These findings suggest that RecCLIP provides a principled foundation for developing scalable and transferable recommender systems. A natural direction for future work is to incorporate available textual information to further enhance RecCLIP. It would also be valuable to explore alternative image representations, ranging from binary to grayscale formats, to capture richer interaction patterns.

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A Dataset

Three real-world recommendation datasets are used to evaluate the performance of RecCLIP and other baselines. The details of these datasets are as follows:

- **LastFM-2K** [4] is a music recommendation dataset of user–artist listening interactions, with play counts serving as implicit feedback for long-term user preferences.
- **MovieLens-1M** [15] is a movie recommendation dataset with explicit user ratings across a broad range of films and genres.
- **Amazon Games** [31] is a video game subset of the Amazon Review dataset containing user–product interactions with both implicit and explicit feedback.

B Baselines

To comprehensively evaluate the performance of our method, we select several highly representative and widely recognized baseline methods as follows:

- **GMF** [17] generalizes matrix factorization with a neural network framework. It uses the element-wise product of user and item embeddings to model their interaction, which captures complex relationships for recommendation.
 - **DLCM** [2] leverages a recurrent neural network and optimizes the model with an attention-based loss function.
 - **BM25** [35] is a classical probabilistic retrieval model for document ranking. In our adaptation to recommendation, we redefine term frequency (TF) and inverse document frequency (IDF) over items: TF is the number of co-occurrence between the user’s historical items and a candidate item, while IDF reflects item rarity (items interacted with by fewer users receive higher IDF).
 - **NGCF** [41] applies graph convolutional networks on user–item bipartite graphs. It propagates embeddings across layers to capture high-order connectivity.
 - **LightGCN** [16] simplifies graph convolution by removing non-linear activations and weight transformations. It only aggregates neighbor embeddings and averages across layers, achieving strong performance with low complexity.
 - **DGCF** [42] models users’ multiple latent interests. It learns several independent intent factors on the user–item graph, allowing more precise recommendations that reflect diverse behaviors.
 - **UltraGCN** [30] reformulates graph convolution as constraint-based optimization within the objective function, effectively approximating high-order connectivity through regularization terms rather than explicit message passing.
 - **RankGPT** [37] formulates the ranking problem as a sequence generation task, leveraging LLMs to directly produce ranked item lists.
 - **PNN** [60] redefines the conventional positive–negative learning paradigm in recommendation by introducing a *neutral* class for ambiguous or unlabeled items. This design establishes a more expressive three-way partial order (positive $\succ$ neutral $\succ$ negative), enabling the model to capture nuanced user preferences and better leverage implicit feedback from unlabeled interactions.
 - **TIGER** [34] pioneers a generative retrieval paradigm by employing a Residual Quantized VAE (RQ-VAE) to map items into hierarchical semantic identifiers. By training a Transformer to autoregressively predict these semantic codes based on interaction history, TIGER effectively captures latent item correlations that distinct ID-based methods often overlook.
- To validate the performance of RecCLIP in the cold start and transferable recommendation scenarios, we select baselines that are specifically designed for these two scenarios respectively.
- **CLCRec** [45] leverages contrastive learning to align item content features with collaborative signals, enhancing representation quality through self-supervised objectives. By bridging semantic and interaction spaces, CLCRec effectively alleviates the cold-start problem for items with limited user interactions and improves recommendation accuracy in sparse data scenarios.
 - **TransRec** [38] relies solely on item textual semantics to model user–item interactions through a tower-based architecture, which discards ID-dependent user and item representations. This design

enables transferable recommendation across domains and maintains robust performance even under sparse or heterogeneous data conditions.

C Implementation Details

In this section, we elaborate the implementation detail in this work. Both H and W are set to 224, following the image resolution used of CLIP and BLIP, respectively. In Section 3.2, we fix the model size throughout all experiments, adopting ViT-B/32 as the vision backbone of the CLIP framework. For GMF, DLCM, NDCG, LightGCN, DGCF and UltraGCN, we set the learning rate at $1e-3$ and weight decay at $1e-4$. For BM25, we set $k1$ at 1.2 and b at 0.75. For RankGPT, GPT-4o-mini [32] is selected as the default LLM backbone. Experiments are conducted on 4 Nvidia 3090 GPUs.

D Complete Experimental Results of Transferable Recommendation

In this section, we present the comprehensive experimental results of transferable recommendation to supplement the analysis in the main text. Table 6 reports the complete performance comparison when models are trained on one source dataset and evaluated on both the source dataset (in-domain) and the target datasets (cross-domain), allowing us to examine the transferability in different perspectives.

For cross-domain performance, RecCLIP demonstrates superior generalization capability when compared with all baseline models. As shown in Table 6, when transferring from MovieLens-1M to LastFM, RecCLIP achieves a Recall@1 of 99.30% and a Normalized Discounted Cumulative Gain (NDCG@1) of 99.30%. This represents a significant improvement over the best-performing baseline, TransRec, which scored 45.23% in both metrics. Similar substantial gains are observed in other transfer scenarios. When transferring from MovieLens-1M to Amazon Games, our model's Recall@1 of 92.67% starkly contrasts with TransRec's 41.76%. This pattern of outperformance is consistent across all evaluation metrics, including Recall and NDCG at various cutoffs (R@1,

R@3, R@5, N@1, N@3, N@5), underscoring the model's enhanced ability to learn transferable representations for users and items that remain effective even when applied to new, unseen domains.

For in-domain performance, while all models experience performance degradation when transferring to new domains, RecCLIP exhibits the smallest performance drop, and in some domains, the metrics even improve when compared with results on the same domain in Table 2. For example, in the transfer from the MovieLens-1M dataset to the LastFM dataset, our model's Recall@1 metric saw an increase. This resilience contrasts sharply with the baseline models, which consistently experienced significant performance drops in cross-domain scenarios. This suggests that the representations learned by our method are more robust and less susceptible to changes in the underlying data characteristics, a crucial feature for building truly domain-agnostic recommendation systems.

The comprehensive results also shed light on the influence of dataset characteristics on transfer learning efficacy. The experiments show that models trained on larger and denser datasets, such as MovieLens-1M, generally yield better performance when transferred to other domains. This observation supports the strategy of leveraging large-scale, information-rich datasets for pre-training models. By doing so, the model can learn more generalized patterns of user preferences and item characteristics, which then serve as a strong foundation for adaptation to more specialized or sparser target domains. This finding has important practical implications for the development of "foundation models" in the field of recommendation systems.

In summary, the experimental evidence strongly supports the conclusion that the proposed method offers a significant advancement in transferable recommendations. Its ability to generalize effectively across diverse domains, its robustness to domain shift, and the clear benefits derived from training on large-scale datasets collectively highlight its potential for a wide range of real-world applications.

Table 6: Performance comparison of transfer learning between our method and baseline methods. All results are presented in percentage format (%). The experiments are conducted by training on one dataset and testing on another, using LastFM, MovieLens-1M, and Amazon Games. For each task (row), the best and second-best results across all methods are highlighted in bold font and underlined, respectively. “*” indicates that our method’s performance is significantly better than all baselines.

Method	Train / Test	R@1	R@3	R@5	N@1	N@3	N@5
NGCF	MovieLens-1M train / LastFM test	23.64	52.63	68.21	23.64	40.24	46.65
lightGCN		23.64	53.59	67.78	23.64	40.97	46.82
DGCF		23.56	54.29	68.30	23.56	41.26	47.02
ultraGCN		22.42	52.45	67.34	22.42	39.92	46.06
TransRec		<u>45.23</u>	<u>68.57</u>	<u>81.32</u>	<u>45.23</u>	<u>56.93</u>	<u>62.68</u>
our method		99.30*	99.91*	99.91*	99.30*	99.67*	99.67*
NGCF	MovieLens-1M train / Amazon Games test	18.31	45.15	61.57	18.31	33.71	40.45
lightGCN		12.96	33.71	48.38	12.96	24.83	30.84
DGCF		12.97	33.85	48.36	12.97	24.90	30.86
ultraGCN		13.20	33.64	48.36	13.20	24.90	30.94
TransRec		41.76	64.14	78.49	41.76	52.68	58.95
our method		92.67*	98.71*	98.96*	92.67*	96.48*	96.62*
NGCF	LastFM train / MovieLens-1M test	28.66	55.39	67.63	28.66	<u>50.25</u>	56.16
lightGCN		16.80	39.07	52.90	16.80	29.59	35.27
DGCF		14.35	36.08	53.13	14.35	26.77	33.75
ultraGCN		16.18	38.34	53.08	16.18	28.88	34.92
TransRec		<u>43.20</u>	<u>59.43</u>	<u>69.34</u>	<u>43.20</u>	50.20	54.35
our method		49.23*	60.69	72.49	49.23*	55.76*	60.60*
NGCF	LastFM train / Amazon Games test	17.92	44.68	64.56	17.92	33.20	41.35
lightGCN		11.36	29.79	46.63	11.36	21.83	28.71
DGCF		9.69	28.21	46.22	9.69	20.17	27.54
ultraGCN		11.36	30.22	46.46	11.36	22.10	28.75
TransRec		<u>38.77</u>	<u>58.29</u>	<u>72.54</u>	<u>38.77</u>	<u>48.69</u>	<u>53.42</u>
our method		57.62*	86.69*	94.84*	57.62*	74.74*	78.12*
NGCF	Amazon Games train / LastFM test	17.51	41.51	61.30	17.51	31.32	39.46
lightGCN		13.66	33.01	49.47	13.66	24.75	31.49
DGCF		18.08	36.85	56.48	18.08	28.89	34.47
ultraGCN		16.02	37.65	56.22	16.02	28.25	35.87
TransRec		<u>39.29</u>	<u>60.52</u>	<u>74.48</u>	<u>39.29</u>	<u>49.76</u>	<u>55.63</u>
our method		81.44*	94.83*	96.76*	81.44*	89.48*	90.27*
NGCF	Amazon Games train / MovieLens-1M test	22.14	49.44	<u>63.74</u>	22.14	37.83	45.45
lightGCN		15.99	37.30	53.36	15.99	28.18	34.77
DGCF		15.76	35.64	49.30	15.76	27.20	32.80
ultraGCN		17.70	42.42	60.96	17.70	31.83	39.43
TransRec		<u>36.63</u>	<u>51.28</u>	61.15	<u>36.63</u>	<u>43.85</u>	<u>51.27</u>
our method		39.94*	58.77*	65.91*	39.94*	51.12*	54.05*